



HITEC UNIVERSITY TAXILA CANTT
BS CYBER SECURITY
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AI LAB REPORT:1

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LAB REPORT: 1

QUESTION NO:1

Write a Python Code to write the table of a number given by the user?

CODE:

```
num = int(input("Enter a number to print its table: "))

print(f"\nMultiplication Table of {num}:\n")
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

OUTPUT:

```
Enter a number to print its table: 7

Multiplication Table of 7:

7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
PS C:\Users\HP> █
```

QUESTION NO:2

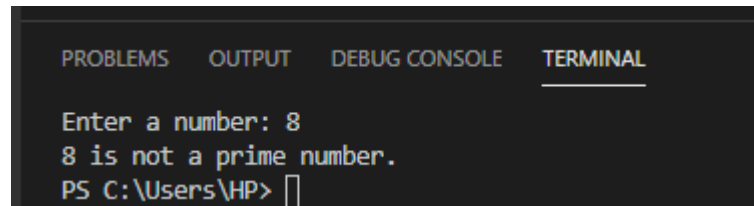
Write a python code to check wheatear a number is prime or not?

CODE:

```
num = int(input("Enter a number: "))

if num > 1:
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            print(f"{num} is not a prime number.")
            break
    else:
        print(f"{num} is a prime number.")
else:
    print(f"{num} is not a prime number.")
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

Enter a number: 8
8 is not a prime number.
PS C:\Users\HP>
```

QUESTION NO:3

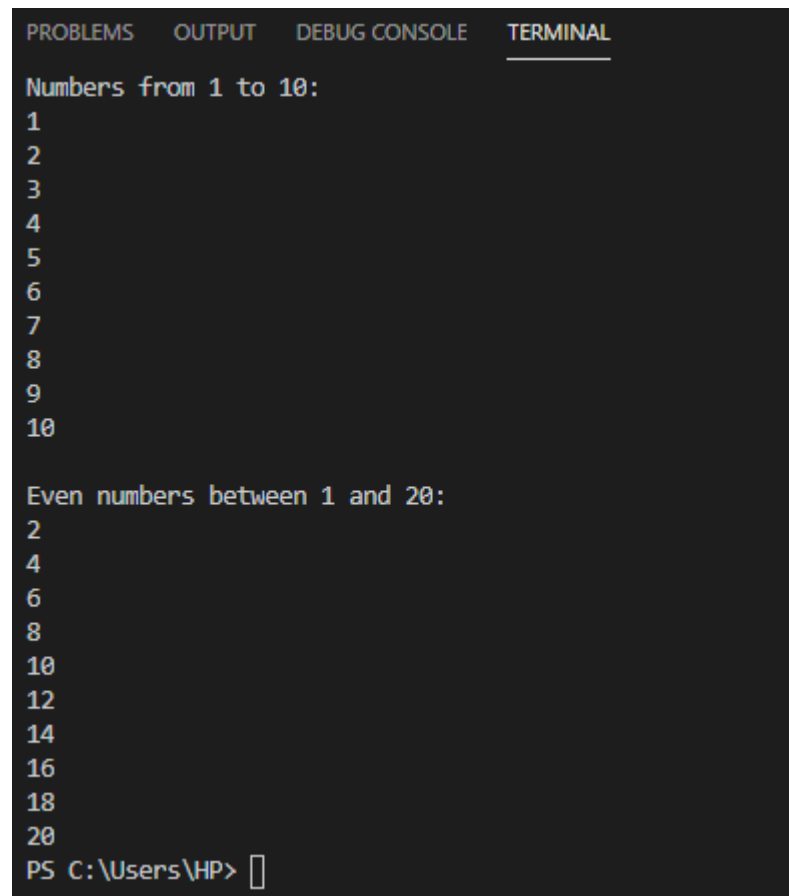
Write a Python program that prints numbers from 1 to 10 using a while loop. Modify the while loop to print only even numbers between 1 and 20?

CODE:

```
print("Numbers from 1 to 10:")
i = 1
while i <= 10:
    print(i)
    i += 1

print("\nEven numbers between 1 and 20:")
i = 2
while i <= 20:
    print(i)
    i += 2
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

Numbers from 1 to 10:
1
2
3
4
5
6
7
8
9
10

Even numbers between 1 and 20:
2
4
6
8
10
12
14
16
18
20
PS C:\Users\HP> 
```

QUESTION NO:4

Write a Python program to determine whether the given number is a Harshad Number. If a number is divisible by the sum of its digits, then it will be known as a Harshad Number?

CODE:

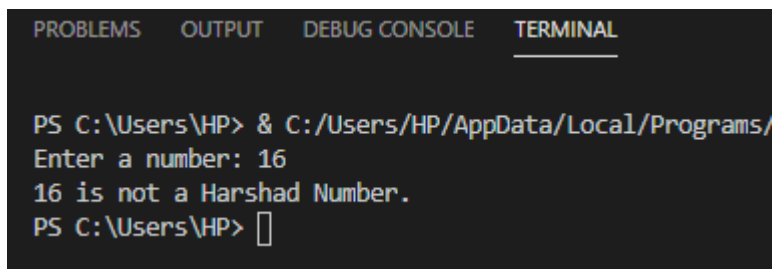
```
num = int(input("Enter a number: "))

sum_of_digits = 0
temp = num

while temp > 0:
    digit = temp % 10
    sum_of_digits += digit
    temp //= 10

if num % sum_of_digits == 0:
    print(f"{num} is a Harshad Number.")
else:
    print(f"{num} is not a Harshad Number.")
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Python.exe
Enter a number: 16
16 is not a Harshad Number.
PS C:\Users\HP> 
```

CONCLUSION:

In this lab, we learned the basics of Python programming, including loops, conditionals, and user input. We practiced programs like printing tables, checking prime and Harshad numbers, and using while loops. This helped us understand logical thinking and how to apply Python for simple mathematical problems.
